

Comparative Mind Map: AI Architecture Landscape

A structural comparison across learning approaches, data requirements, complexity, outputs, and educational applicability

Artificial Intelligence (AI)	Machine Learning (ML)	Deep Learning (DL)	Generative AI (GenAI)	Large Language Models
BROADEST CONCEPT	SUBSET OF AI	SUBSET OF ML	DL CAPABILITY BRANCH	GENAI TEXT SPECIALIZATION
LEARNING APPROACH Rule-based systems, deterministic logic, and basic heuristics. Mimics human decisions via hardcoded logic.	LEARNING APPROACH Supervised, unsupervised, & reinforcement learning. Parses data to discover underlying patterns.	LEARNING APPROACH Multi-layered Artificial Neural Networks. Automatically extracts intricate high-level features from raw data.	LEARNING APPROACH Generative Adversarial Networks (GANs) / Diffusion systems. Learns data distributions to build new instances.	LEARNING APPROACH Transformer models utilizing self-attention mechanism. Pre-trained on token sequence forecasting.
DATA REQUIREMENTS Minimal to none for expert systems; relies heavily on pre-defined structures rather than continuous datasets.	DATA REQUIREMENTS High. Requires thousands of clean, structured records (Excel/CSV formats) for training accuracy.	DATA REQUIREMENTS Very High. Demands massive unstructured datasets (millions of images, audio clips, text logs).	DATA REQUIREMENTS Comprehensive. Requires enormous multimodal datasets spanning millions of text-image-audio pairs.	DATA REQUIREMENTS Web-scale text corpora. Ingests massive libraries, code repositories, websites, and encyclopedias.
COMPUTE COMPLEXITY Low to Moderate. Executes fast logical tree evaluations with highly predictable resource usage.	COMPUTE COMPLEXITY Moderate. Requires CPU/GPU clusters for model training, but real-time inference runs rapidly.	COMPUTE COMPLEXITY High to Severe. Requires heavy GPU/TPU acceleration for complex matrix backpropagation loops.	COMPUTE COMPLEXITY Astronomical. Massive multi-cluster data center setups needed for baseline model weights.	COMPUTE COMPLEXITY Astronomical. Highly resource-demanding token-by-token generation phases requiring persistent memory.
PRIMARY OUTPUTS Discrete decisions, basic structured pathfinding, rules-based responses.	PRIMARY OUTPUTS Numerical predictions, regression trends, item classifications, data clustering.	PRIMARY OUTPUTS Deep feature representations, complex object classification maps, audio transcripts.	PRIMARY OUTPUTS Entirely new synthetic content (images, synthesized voice tracks, videos, 3D structures).	PRIMARY OUTPUTS Context-aware text streams, structural
		KEY APPLICATIONS		

KEY APPLICATIONS

Traditional robotics, expert tax preparation software, basic chess engines.

KEY APPLICATIONS

Fraud detection filters, retail recommendation engines, spam filters.

Biometric face recognition, self-driving situational vision, speech-to-text.

KEY APPLICATIONS

Midjourney art engines, dynamic video synthesis, artificial voice clones.

program code, translated script, reasoned logic.

KEY APPLICATIONS

Advanced conversational tools (ChatGPT, Claude), automated code assistants, contextual search.

Strategic Technology Recommendation for Educational Chatbots

Primary Choice: Large Language Models (LLMs) combined with **Retrieval-Augmented Generation (RAG)**.

Traditional AI/ML models fail to process the semantic nuance of unique student questions. LLMs simulate an adaptable, interactive human tutor capable of reframing definitions iteratively based on student prompts.

1.

Conversational Adaptability: Handles complex multi-turn dialogs seamlessly, enabling a supportive Socratic learning method.

2. Safety & Precision via RAG: Overcomes standard LLM hallucinations by anchoring responses explicitly to verified course syllabi, textbooks, and guidelines.

3. Multidisciplinary Scope: Switches instantly from resolving algebraic equations to explaining historical timelines within a single user window.